Change Request Log & Impact Assessment — WariMitra

Document Identifier: WM-CR-019

Classification: Agile Change Control & Requirements Evolution

Standard: BABOK v3 Chapter 5 (Requirements Life Cycle Management - Assess Requirements Changes)

Rule of Representation: Every simulated scenario is explicitly tagged as `[SIMULATED INTERVIEW SCENARIO]` to distinguish it from the codebase baseline.

1. Change Management Process & Governance

When a stakeholder requests a scope modification, the Business Analyst leads the Change Control process using a 5-step framework:

1. **Intake & Logging:** Capture proposed change in the Change Request Log.

2. **Impact Assessment:** Evaluate impact across Triple Constraints (Scope, Time/Schedule, Budget/Cost), plus Architecture, Security, and Compliance.

3. **Change Control Board (CCB) Review:** Present findings to Project Sponsor, Lead Architect, and District Police/Medical representatives.

4. **Disposition:** Formal outcome: `Approved`, `Rejected`, or `Deferred to Backlog`.

5. **Traceability Update:** If approved, update BRD, FRD, User Stories, and RTM.

2. Change Request Log

CR ID	Title / Request Summary	Requesting Stakeholder	Business Justification	Architectural & Technical Impact	Estimated Effort	CCB Decision	Implementation Target
CR-001 `[ACTUAL HISTORICAL EVOLUTIO`	**Pivot from Monolithic Django**	Lead Solution Architect & Senior	Monolithic Django 6.0 with synchrono	Complete rewrite of backend to	4 Weeks	**Approved**	v2.0 Production Cutover (`ARCHITEC`

EVOLUTIO	**c Django**	& Senior	synchrono	to			(ARCHITEC

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N]	**to Firebase Serverless v2.0**	BA	us DB writes could not support real-time GPS pings from 500k+ pilgrims without fatal database lockups.	Firebase Cloud Functions and Cloud Firestore (asia-south1).			TURE.md:553)
CR-002 [SIMULATED INTERVIEW SCENARIO]	**Add Aadhaar-based Face Authentication for Darshan Slot Booking**	Temple Trust Executive	Reduce scalping and ensure only the registered devotee enters the VIP queue.	High technical complexity; requires integration with UIDAI Aadhaar biometric eKYC gateway; data privacy and DPDP Act	6 Weeks	**Rejected**	Rejected due to privacy risk and network instability at rural gates.

CR ID	Title / Request Summary	Requesting Stakeholder	Business Justification	Architectural & Technical Impact	Estimated Effort	CCB Decision	Implementation Target
				Act compliance liabilities.			
CR-003 `[SIMULATED INTERVIEW SCENARIO]`	**Automated Multi-lingual Voice Alert Dispatch (Marathi IVR)**	Rural Dindi Leaders Association	Many elderly devotees (75+ years) cannot read text notifications or hear phone pings in loud cymbal-playing processions.	Integration with an external Twilio / Exotel telephony gateway to trigger outbound automated phone calls with pre-recorded Marathi voice.	2 Weeks	**Approved**	Deferred to Post-Wari v2.1
CR-004 `[SIMULATED INTERVIEW SCENARIO]`	**Dynamic Water Tanker Telemetry Integration (IoT Flow Meters)**	Pune Municipal Corporation Water Works	Ensure water tankers along the highway are refilled before running completel	Ingest ultrasonic liquid level sensor pings via MQTT into a new Firestore	3 Weeks	**Approved**	Slated for Phase 3 Expansion

			completely dry during midday heat.	Firestore collection `waterTankers`.			

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3. Deep-Dive Change Impact Analysis: Example CR-001 (Django to Firebase Pivot)

3.1 Business Context

During Stage 1 (Varithon Hackathon), the initial backend was implemented as a Django 6.0 application backed by SQLite/PostgreSQL. An architectural pre-production review (`Reference files/backend_audit.md`) evaluated this architecture and issued a score of 6.2/10 (High Risk), warning that synchronous GPS updates and WebSocket broadcasting would exhaust database connections and CPU.

3.2 Impact Analysis Conducted by the Business Analyst:

1. **Scope Impact:** Removed relational migrations and Django Channels; adopted serverless NoSQL documents and native real-time listeners.
2. **Performance Impact:** Reduced SOS ingestion latency from projected 8–15s under peak queue congestion to an empirical 1.2s p95 on Cloud Functions.
3. **Cost Impact:** Eliminated fixed monthly EC2/RDS server costs ($400+/mo), moving to Firebase Blaze pay-as-you-go projected at ~$185/month for 1M users.
4. **Risk Mitigation:** Decoupled client operations from server maintenance windows; native offline persistence in Flutter app ensured field resilience.